

MAT 2305- Real Analysis II
Assignment # 02

Attain all the questions and be prepared to submit solutions for the following questions on or before **9th December, 2024**.

$$1[(i), (iv)], 2, 3, 4[(i), (iv)], 6.$$

1. Find the interval of convergence of each of the following power series:

- i. $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$
- ii. $\sum_{n=0}^{\infty} ((-1)^n + 3)^n (x - 1)^n$
- iii. $\sum_{n=1}^{\infty} \log\left(\frac{n+1}{n}\right) x^n$
- iv. $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdots (2n-1)}{(k+1)!} x^{2n}$
- v. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x - 1)^n$
- vi. $\frac{x}{2} + \frac{1}{2} \cdot \frac{3}{5} x^2 + \frac{1}{2} \cdot \frac{3}{5} \cdot \frac{5}{8} x^3 + \cdots$

2. Find the radius of convergence and the exact interval of convergence of the power series

$$\sum_{n=0}^{\infty} \frac{(n+1)}{(n+2)(n+3)} x^n.$$

3. Find the radius of convergence and the exact interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n.$$

4. Find a closed form for each of the following series and the largest set on which this formula is valid. (Hint: A closed form for the power series $\sum_{n=0}^{\infty} x^n = \frac{1}{x-1}$; $|x| < 1$)

- i. $\sum_{n=1}^{\infty} 3x^{3n-1}$
- ii. $\sum_{n=2}^{\infty} nx^{n-2}$
- iii. $\sum_{n=1}^{\infty} \frac{2n}{n+1} (1-x)^n$
- iv. $\sum_{n=0}^{\infty} \frac{x^{3n}}{n+1}$

5. Show that the radius of convergence, R , of power series $\sum a_n(x - c)^n$ is given by

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|,$$

provided that this limit exists.

6. Show that

$$\ln(x + 1) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

, where $-1 \leq x \leq 1$.

Deduce that

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

7. Show by integrating the power series of $\frac{1}{1+x}$; $|x| < 1$,

$$\ln(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n.$$

8. Show that

$$\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots; -1 \leq x \leq 1.$$

9. Show that

$$\sin^{-1}x = x + \frac{1}{2 \cdot 3}x^3 + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5}x^5 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 7}x^7 + \dots; -1 < x \leq 1.$$

10. If the power series $\sum_{n=1}^{\infty} a_n x^n$ has the radius of convergence R and $a_n \neq 0$ for large n , show that

$$\liminf_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \leq R \leq \limsup_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|.$$